

Limits of functions and asymptotes



Outline

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Sequential definition of limit



Introduction: sequential definition of limit

- ▶ The limit operation can be extended from sequences to real functions of a real variable.
- ▶ In this way we can describe the behavior of the function when the independent variable **moves close to a given point** or **becomes very large** (in absolute value).
- ▶ The sequential definition reduces the limit of a function to the **limit of a sequence**, which we have already studied

The setting

- ▶ Consider an **interval** I , a **point** $c \in I$ and a **function** f defined in I , except possibly at the point c .
- ▶ The **interval** I can be **bounded** or **unbounded**, **closed** or **open**.
- ▶ The **point** c can be **interior** to the interval or **one of its endpoints** (possibly $+\infty$ or $-\infty$).

Sequential definition of limit

Definition 1: Limit (sequential definition)

With $\ell, c \in \mathbb{R}^*$, we write

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c$$

if for every sequence $\{x_n\}$ of points of I such that $x_n \rightarrow c$ as $n \rightarrow +\infty$ and $x_n \neq c, \forall n$, we have

$$f(x_n) \rightarrow \ell \text{ as } n \rightarrow +\infty$$

In these cases we can also write:

$$\lim_{x \rightarrow c} f(x) = \ell$$

which is read: the limit of $f(x)$, as x tends to c , is ℓ .

Interpretation

- ▶ In other words, if **for any sequence $\{x_n\}$ of inputs** the **sequence of outputs $\{f(x_n)\}$** tends to the limit ℓ (finite or infinite), then we say that the limit of $f(x)$, as x tends to c , is ℓ .
- ▶ The sequential definition of limit reduces the concept of limit of a function to that of limit of a sequence.
- ▶ **It is not the only possible definition.** Another definition is the *topological definition of limit*. The two definitions are perfectly equivalent.

Sequential definition of limit:

- ▶ $f(x) \rightarrow \ell$ as $x \rightarrow c$ if for every sequence $x_n \rightarrow c$, with $x_n \neq c$, we have $f(x_n) \rightarrow \ell$
- ▶ c and ℓ can be finite or infinite ($c, \ell \in \mathbb{R}^*$)
- ▶ The limit of a function is reduced to the limit of a sequence
- ▶ There is an equivalent definition: the topological definition

Neighborhoods and topological definition of limit



Introduction: neighborhoods and topological definition of limit

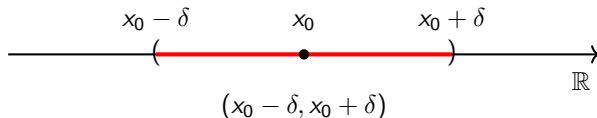
- ▶ In this section we introduce the **neighborhoods** of a point and of $\pm\infty$
- ▶ With neighborhoods we define the expression **eventually** and the **topological definition of limit**, equivalent to the sequential one
- ▶ We will also see the **uniqueness of the limit theorem** and the notions of **infinitesimal** and **infinite** function

Centered neighborhood of a point

Definition 2: Centered neighborhood of a point

Given $\delta > 0$ and $x_0 \in \mathbb{R}$, the **neighborhood** centered at x_0 associated with δ is the open interval:

$$(x_0 - \delta, x_0 + \delta) = \{x \in \mathbb{R} : x_0 - \delta < x < x_0 + \delta\}$$



Moving in a neighborhood

Saying that “ x moves in a neighborhood of x_0 ” means stating that

$$x \in (x_0 - \delta, x_0 + \delta), \text{ where we think of } 0 < \delta \ll 1.$$

Neighborhood of $\pm\infty$

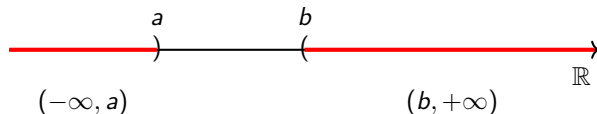
Definition 3: Neighborhood of $\pm\infty$

Given $a, b \in \mathbb{R}$, the neighborhood of $-\infty$ associated with a is the interval:

$$(-\infty, a) = \{x \in \mathbb{R} : x < a\}$$

and the neighborhood of $+\infty$ associated with b is the interval:

$$(b, +\infty) = \{x \in \mathbb{R} : x > b\}$$



Property eventually satisfied

- We now introduce the expression “eventually as $x \rightarrow c$ ”, analogously to what we did in the case of sequences

Definition 4: Property eventually satisfied by functions

*We say that a function f **eventually** has a certain property as $x \rightarrow c \in \mathbb{R}^*$ if there exists a neighborhood U of c such that the property holds for $f(x)$ for every $x \in U, x \neq c$.*

Topological definition of limit

- We now give the topological definition of limit which, as already mentioned, is equivalent to the sequential one seen above, but turns out to be useful.

Definition 5: Limit (topological definition)

Let $c, \ell \in \mathbb{R}^$, and let f be a function defined at least eventually as $x \rightarrow c$. We write*

$$f(x) \rightarrow \ell \quad \text{as} \quad x \rightarrow c$$

if for every neighborhood U_ℓ of ℓ there exists a neighborhood U_c of c such that

$$\forall x \in U_c, x \neq c, \quad \text{we have} \quad f(x) \in U_\ell$$

The specific cases (1/2)

This definition can be specialized to the specific cases:

Finite limit at a finite point: $f(x) \rightarrow \ell \in \mathbb{R}$ as $x \rightarrow c \in \mathbb{R}$

$$\forall \varepsilon > 0, \exists \delta > 0 : \quad \forall x \neq c, |x - c| < \delta \implies |f(x) - \ell| < \varepsilon$$

Infinite limit at a finite point: $f(x) \rightarrow \pm\infty$ as $x \rightarrow c \in \mathbb{R}$

$$\forall M > 0, \exists \delta > 0 : \quad \forall x \neq c, |x - c| < \delta \implies |f(x)| > M$$

The specific cases (2/2)

Finite limit at infinity: $f(x) \rightarrow \ell \in \mathbb{R}$ as $x \rightarrow \pm\infty$

$$\forall \varepsilon > 0, \exists b > 0 : \quad \forall |x| > b \implies |f(x) - \ell| < \varepsilon$$

Infinite limit at infinity: $f(x) \rightarrow \pm\infty$ as $x \rightarrow \pm\infty$

$$\forall M > 0, \exists b > 0 : \quad \forall |x| > b \implies |f(x)| > M$$

Why the sequential definition

The fact that we have already developed the basics of *limit calculus for sequences* will make it very convenient to use the sequential definition of limit to prove the theorems on limits of functions starting from those seen for sequences.



Uniqueness of the limit

Theorem 1: Uniqueness of the limit of functions

Given a function f , if $f(x) \rightarrow \ell$ as $x \rightarrow c$, then this limit ℓ is unique.

Proof

If there were two limits ℓ_1 and ℓ_2 , different from each other, taking any sequence $\{x_n\}$ such that $x_n \rightarrow c$ and $x_n \neq c, \forall n$, we would have:

$$f(x_n) \rightarrow \ell_1 \quad \text{and} \quad f(x_n) \rightarrow \ell_2 \quad \text{as} \quad n \rightarrow +\infty$$

Hence the sequence $\{f(x_n)\}$ would have two distinct limits: a contradiction. □

Infinitesimal and infinite functions

Definition 6: Infinitesimal function

A function f such that $f(x) \rightarrow 0$ as $x \rightarrow c$ is called **infinitesimal** as $x \rightarrow c$.

Definition 7: Infinite function

A function f such that $f(x) \rightarrow \pm\infty$ as $x \rightarrow c$ is called **infinite** as $x \rightarrow c$.

Finite/infinite limits, at a finite point/at infinity

When we write:

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c \quad \text{or} \quad \lim_{x \rightarrow c} f(x) = \ell$$

we have

$$\text{a } \begin{cases} \text{finite limit} \\ \text{infinite limit} \end{cases} \quad \text{if} \quad \begin{cases} \ell \in \mathbb{R} \\ \ell = \pm\infty \end{cases}$$

we have

$$\text{a limit } \begin{cases} \text{at a finite point} \\ \text{at infinity} \end{cases} \quad \text{if} \quad \begin{cases} c \in \mathbb{R} \\ c = \pm\infty \end{cases}$$

Summary

Neighborhoods:

- ▶ Neighborhood of x_0 : $(x_0 - \delta, x_0 + \delta)$; neighborhoods of $-\infty$ and $+\infty$: $(-\infty, a)$ and $(b, +\infty)$
- ▶ “Eventually as $x \rightarrow c$ ”: in a neighborhood of c , possibly excluding c

Topological definition of limit:

- ▶ For every neighborhood of ℓ there is a neighborhood of c whose images (for $x \neq c$) lie in the neighborhood of ℓ
- ▶ Four cases: finite/infinite, at a finite point/at infinity
- ▶ The limit, if it exists, is **unique**; **infinitesimal** ($\ell = 0$) and **infinite** ($\ell = \pm\infty$) functions

Finite limit at infinity

Introduction: finite limit at infinity

- In this section we study the case

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c \quad \text{with} \quad \ell \in \mathbb{R} \quad \text{and} \quad c = \pm\infty$$

- We will see how to use the sequential definition to prove a limit of this type
- Graphically, this case corresponds to a **horizontal asymptote**; we will also see when the limit is **from above** or **from below**

Example (1/2)

Let us prove that:

$$e^x \rightarrow 0 \quad \text{as} \quad x \rightarrow -\infty$$

By the definition, we have to prove that for every sequence $\{x_n\}$ such that $x_n \rightarrow -\infty$ as $n \rightarrow +\infty$, we have:

$$e^{x_n} \rightarrow 0 \quad \text{as} \quad n \rightarrow +\infty$$

By the definition of limit of a sequence, this means proving that, for every $\varepsilon > 0$, we have:

$$|e^{x_n}| < \varepsilon, \quad \text{eventually,} \quad \text{that is} \quad e^{x_n} < \varepsilon, \quad \text{eventually,}$$

since the exponential is always positive.

Example (2/2)

The last inequality is equivalent to:

$$x_n < \log \varepsilon$$

If $\varepsilon > 0$ is a small number (< 1), $\log \varepsilon$ is a negative number (and large in absolute value). We set

$$M(\varepsilon) = -\log \varepsilon > 0$$

We therefore have to prove that, for every $M(\varepsilon) > 0$, we have:

$$x_n < -M(\varepsilon), \text{ eventually.}$$

But this is exactly what holds by hypothesis, because $x_n \rightarrow -\infty$.
Hence $e^x \rightarrow 0$ as $x \rightarrow -\infty$.

Horizontal asymptote

Definition 8: Horizontal asymptote

We say that f has a **horizontal asymptote** with equation $y = \ell \in \mathbb{R}$ as $x \rightarrow +\infty$ ($x \rightarrow -\infty$) if:

$$f(x) \rightarrow \ell \in \mathbb{R} \quad \text{as } x \rightarrow +\infty \quad (x \rightarrow -\infty)$$

- Every situation of *finite limit at infinity*, therefore, corresponds graphically to the **presence of a horizontal asymptote**, that is, a horizontal line which the graph of the function gets closer and closer to.

Limit from above or from below

- ▶ When a function has a finite limit at infinity, it is **sometimes** possible to specify whether this limit is **from above** (ℓ^+) or **from below** (ℓ^-). Graphically, this means that the graph of the function approaches the level $y = \ell$ from **above** or from **below**.

Definition 9: Limit from above or from below

With $\ell, c \in \mathbb{R}^$, we write*

$$f(x) \rightarrow \ell^+ \quad (\ell^-) \quad \text{as } x \rightarrow c$$

if for every sequence $\{x_n\}$ of points of I such that $x_n \rightarrow c$ as $n \rightarrow +\infty$ and $x_n \neq c, \forall n$, we have

$$f(x_n) \rightarrow \ell^+ \quad (\ell^-) \quad \text{as } n \rightarrow +\infty$$

Meaning

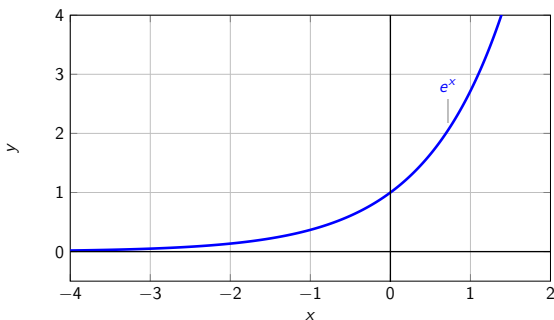
- ▶ Stating that $f(x) \rightarrow \ell^+$ means that $f(x) \rightarrow \ell$ and moreover $f(x) \geq \ell$, eventually.
- ▶ Stating that $f(x) \rightarrow \ell^-$ means that $f(x) \rightarrow \ell$ and moreover $f(x) \leq \ell$, eventually.
- ▶ In these cases we say that $f(x)$ tends to ℓ from above (from below) as x tends to c .

Example

For example:

$$\lim_{x \rightarrow -\infty} e^x = 0^+$$

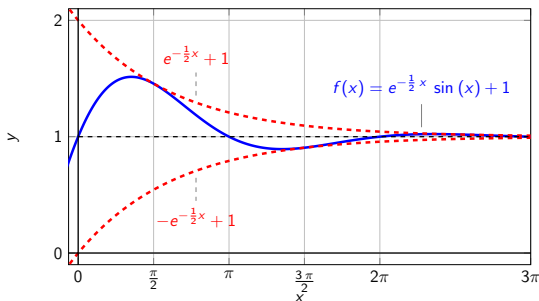
the notation 0^+ means that the function tends to 0 from above, that is, the values of $f(x)$ tend to zero while remaining non-negative.



Example

Note that not every finite limit is necessarily from above or from below, as shown for example by:

$$\lim_{x \rightarrow +\infty} e^{-\frac{1}{2}x} \sin(x) + 1 = 1$$



The function $f(x)$ tends to 1 as $x \rightarrow +\infty$, but we can state neither $f(x) \rightarrow 1^+$ nor $f(x) \rightarrow 1^-$.

Summary

Finite limit at infinity:

- ▶ $f(x) \rightarrow \ell \in \mathbb{R}$ as $x \rightarrow \pm\infty$; example: $e^x \rightarrow 0$ as $x \rightarrow -\infty$
- ▶ It corresponds graphically to a **horizontal asymptote** $y = \ell$

From above or from below:

- ▶ $f(x) \rightarrow \ell^+$ (ℓ^-): $f(x) \rightarrow \ell$ and $f(x) \geq \ell$ ($f(x) \leq \ell$) eventually
- ▶ Not every finite limit is from above or from below

Infinite limit at infinity

Introduction: infinite limit at infinity

- ▶ In this section we study the case

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c \quad \text{with} \quad \ell = \pm\infty \quad \text{and} \quad c = \pm\infty$$

- ▶ We prove a limit of this type with the sequential definition
- ▶ We then introduce the **oblique asymptote** and an operational criterion to find it

Example

Let us prove that:

$$\log_{1/2} x \rightarrow -\infty \quad \text{as} \quad x \rightarrow +\infty$$

By the definition, we have to prove that for every sequence $\{x_n\}$ such that $x_n \rightarrow +\infty$ as $n \rightarrow +\infty$, we have:

$$\log_{1/2} x_n \rightarrow -\infty \quad \text{as} \quad n \rightarrow +\infty$$

This means proving that for every $M > 0$ we have:

$$\log_{1/2} x_n < -M, \quad \text{eventually, that is}$$

$$x_n > \left(\frac{1}{2}\right)^{-M} = 2^M, \quad \text{eventually}$$

But by hypothesis $x_n \rightarrow +\infty$, so, having fixed the positive quantity 2^M , we certainly have $x_n > 2^M$, eventually. Hence $\log_{1/2} x \rightarrow -\infty$ as $x \rightarrow +\infty$.

Oblique asymptote

- ▶ When a function has an infinite limit at infinity, it may happen (but it does not always happen) that there exists an oblique line which the graph of the function gets closer and closer to.

Definition 10: Oblique asymptote

We say that f has an **oblique asymptote** with equation $y = mx + q$ ($q, m \in \mathbb{R}, m \neq 0$) as $x \rightarrow +\infty$ ($-\infty$) if:

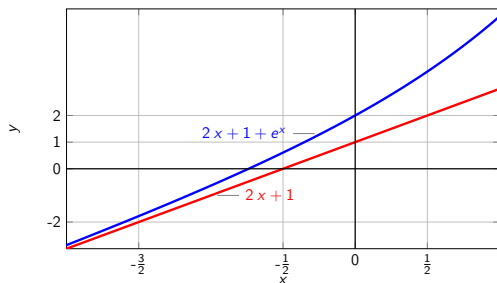
$$(f(x) - (mx + q)) \rightarrow 0 \quad \text{as } x \rightarrow +\infty \quad (-\infty)$$

Example

Consider $f(x) = 2x + 1 + e^x$ and we want to test whether or not $2x + 1$ is an oblique asymptote as $x \rightarrow -\infty$. We have

$$\lim_{x \rightarrow -\infty} (f(x) - (2x + 1)) = \lim_{x \rightarrow -\infty} e^x = 0$$

Therefore, by the definition of oblique asymptote, the line $y = 2x + 1$ is an oblique asymptote of f as $x \rightarrow -\infty$.



Existence of an oblique asymptote

- In less elementary cases, instead of having to “guess” the oblique asymptote, it is useful to have an operational criterion to find it.

Proposition 1: Existence of an oblique asymptote

A function f has an oblique asymptote with equation $y = mx + q$ as $x \rightarrow +\infty$ if and only if the following two conditions hold:

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = m \in \mathbb{R}, m \neq 0 \quad \text{and}$$

$$\lim_{x \rightarrow +\infty} (f(x) - mx) = q \in \mathbb{R}$$

An analogous criterion holds for $x \rightarrow -\infty$.

Remarks on the criterion

- Note that the first condition requires $f(x)$ to have the same order of infinity as $y = x$ as $x \rightarrow +\infty$, since:

$$\lim_{x \rightarrow -\infty} x = -\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} x = +\infty$$

Moreover, if $m > 0$ we have:

$$\lim_{x \rightarrow -\infty} f(x) = -\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} f(x) = +\infty$$

and if $m < 0$ we have:

$$\lim_{x \rightarrow -\infty} f(x) = +\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} f(x) = -\infty$$

The oblique asymptote $y = mx + q$ is a (non-horizontal) line that approximates the behavior of a function that diverges as $x \rightarrow \pm\infty$.

Proof

The first condition rules out the possibility that the asymptote is horizontal and requires $f(x)$ to have the same order of infinity as $y = x$. From the second condition:

$$\text{if } \lim_{x \rightarrow +\infty} (f(x) - mx) = q \quad \text{then} \quad \lim_{x \rightarrow +\infty} (f(x) - (mx + q)) = 0$$

Analogous reasoning for $x \rightarrow -\infty$.



Example

Consider

$$f(x) = 3x + \sqrt{x}$$

and compute

$$\lim_{x \rightarrow +\infty} \frac{3x + \sqrt{x}}{x} = \lim_{x \rightarrow +\infty} \left(3 + \frac{1}{\sqrt{x}} \right) = 3$$

$$\lim_{x \rightarrow +\infty} (3x + \sqrt{x} - 3x) = \lim_{x \rightarrow +\infty} \sqrt{x} = +\infty$$

since the second limit is infinite, the second condition of the proposition does not hold and consequently the function has no oblique asymptote as $x \rightarrow +\infty$.

Summary

Infinite limit at infinity:

- ▶ $f(x) \rightarrow \pm\infty$ as $x \rightarrow \pm\infty$; example: $\log_{1/2} x \rightarrow -\infty$ as $x \rightarrow +\infty$

Oblique asymptote $y = mx + q$, $m \neq 0$:

- ▶ Definition: $f(x) - (mx + q) \rightarrow 0$ as $x \rightarrow \pm\infty$
- ▶ Criterion: $\frac{f(x)}{x} \rightarrow m \in \mathbb{R} \setminus \{0\}$ and $f(x) - mx \rightarrow q \in \mathbb{R}$
- ▶ If the second limit is infinite, there is no oblique asymptote

Infinite limit at a finite point

Introduction: infinite limit at a finite point

- ▶ In this section we study the case

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c \text{ with } \ell = \pm\infty \text{ and } c \in \mathbb{R}$$

- ▶ Here the condition $x_n \neq c$ of the definition becomes relevant
- ▶ We introduce the **right** and **left limits** and the **vertical asymptote**

Example (1/2)

Let us prove that:

$$\frac{1}{x^2} \rightarrow +\infty \quad \text{as} \quad x \rightarrow 0$$

By the definition, we have to prove that for every sequence $\{x_n\}$ such that $x_n \rightarrow 0$ as $n \rightarrow +\infty$ and $x_n \neq 0, \forall n$, we have:

$$\frac{1}{x_n^2} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty$$

Note that the condition $x_n \neq 0, \forall n$, contained in the definition of limit, was irrelevant in the previous examples where we had $x \rightarrow \pm\infty$, but now it clearly becomes relevant.

Example (2/2)

We therefore have to prove that for every $M > 0$ we have:

$$\frac{1}{x_n^2} > M, \quad \text{eventually, that is}$$

$$|x_n| < \frac{1}{\sqrt{M}}, \quad \text{eventually}$$

But by hypothesis $x_n \rightarrow 0$, so, having fixed the positive quantity $\frac{1}{\sqrt{M}}$, we certainly have $|x_n| < \frac{1}{\sqrt{M}}$ eventually. Hence $1/x^2 \rightarrow +\infty$ as $x \rightarrow 0$.

Right and left limits

- Sometimes a function behaves differently, as far as its limit is concerned, depending on whether x approaches c from the **right** or from the **left**.

Definition 11: Right and left limits

With $\ell \in \mathbb{R}^$ and $c \in \mathbb{R}$, we write*

$$f(x) \rightarrow \ell \quad \text{as } x \rightarrow c^+ \quad (c^-)$$

if for every sequence $\{x_n\}$ of points of I such that $x_n \rightarrow c^+ (c^-)$ as $n \rightarrow +\infty$ and $x_n \neq c^+ (c^-)$, $\forall n$, we have

$$f(x_n) \rightarrow \ell \quad \text{as } n \rightarrow +\infty$$

Existence of the limit and right and left limits

The limit $\lim_{x \rightarrow c} f(x)$ exists if and only if the right limit and the left limit exist and are both equal to ℓ . That is, if and only if:

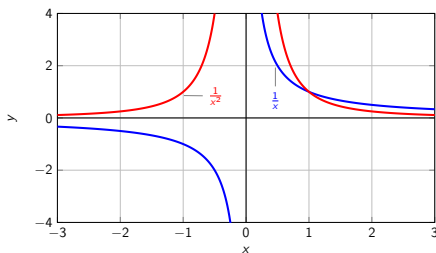
$$\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = \ell$$

However, it may happen that the right limit and the left limit exist but are different from each other, or that only one of the two exists. In these cases:

$$\lim_{x \rightarrow c} f(x) \quad \text{does not exist}$$

Example

Consider the functions $y = \frac{1}{x}$ and $y = \frac{1}{x^2}$:



$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty, \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty \quad \text{while} \quad \lim_{x \rightarrow 0} \frac{1}{x} \text{ does not exist}$$

$$\lim_{x \rightarrow 0^+} \frac{1}{x^2} = +\infty, \quad \lim_{x \rightarrow 0^-} \frac{1}{x^2} = +\infty \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$$

Limits at the endpoints of an interval

If f is defined in (a, b) , the limit as $x \rightarrow a$ (respectively $x \rightarrow b$) is automatically a right limit (respectively a left limit).

Vertical asymptote

Definition 12: Vertical asymptote

We say that f has a **vertical asymptote** with equation $x = c \in \mathbb{R}$ as $x \rightarrow c$ (c^+ or c^-) if:

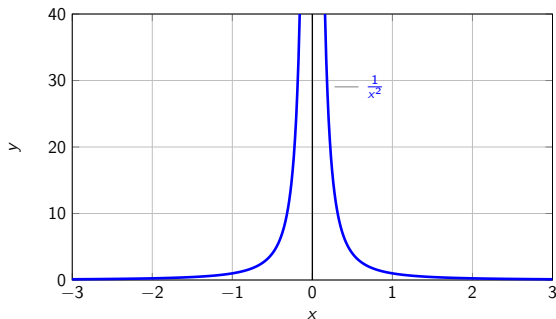
$$f(x) \rightarrow \pm\infty \text{ as } x \rightarrow c \text{ } (c^+ \text{ or } c^-)$$

- ▶ Every situation of **infinite limit at a finite point**, therefore, corresponds graphically to the presence of a **vertical asymptote**, that is, a **vertical line** which the graph of the function gets closer and closer to.

Example (1/2)

$x = 0$ is a vertical asymptote of

$$\frac{1}{x^2} \quad \text{as } x \rightarrow 0$$



Example (2/2)

- ▶ $x = 0$ is a vertical asymptote of

$$\frac{1}{x} \quad \text{as } x \rightarrow 0^+ \text{ and as } x \rightarrow 0^-$$

- ▶ $x = 0$ is a vertical asymptote of

$$\log x \quad \text{as } x \rightarrow 0^+$$

Summary

Infinite limit at a finite point:

- ▶ $f(x) \rightarrow \pm\infty$ as $x \rightarrow c \in \mathbb{R}$; example: $\frac{1}{x^2} \rightarrow +\infty$ as $x \rightarrow 0$
- ▶ The condition $x_n \neq c$ becomes essential

Right and left limits:

- ▶ The limit as $x \rightarrow c$ exists if and only if the right and left limits exist and coincide

Vertical asymptote:

- ▶ $x = c$ is a vertical asymptote if $f(x) \rightarrow \pm\infty$ as $x \rightarrow c$ (or c^+ , or c^-)

Finite limit at a finite point



Introduction: finite limit at a finite point

- ▶ In this section we study the case

$$f(x) \rightarrow \ell \text{ as } x \rightarrow c \quad \text{with} \quad \ell \in \mathbb{R} \quad \text{and} \quad c \in \mathbb{R}$$

- ▶ We will see two examples: in one the limit coincides with the value of the function at the point, in the other it does not
- ▶ This difference will lead us to the concept of **continuity**

Example

Consider the function $f(x) = \sin x$ and let us prove that:

$$\sin x \rightarrow 0 \quad \text{as} \quad x \rightarrow 0$$

By the definition, we have to prove that for every sequence $\{x_n\}$ such that $x_n \rightarrow 0$ as $n \rightarrow +\infty$ and $x_n \neq 0, \forall n$, we have:

$$\sin x_n \rightarrow 0 \quad \text{as} \quad n \rightarrow +\infty$$

We start from the following elementary inequality:

$$|\sin x| \leq |x|, \quad \forall x \in \mathbb{R} \quad \text{hence} \quad |\sin x_n| \leq |x_n|$$

By the comparison theorem:

if $x_n \rightarrow 0$ as $n \rightarrow +\infty$ then also $\sin x_n \rightarrow 0$ as $n \rightarrow +\infty$

Hence $\sin x \rightarrow 0$ as $x \rightarrow 0$ and we have: $\lim_{x \rightarrow 0} \sin x = f(0)$

Example

Consider the function:

$$f(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and let us prove that: $f(x) \rightarrow 1$ as $x \rightarrow 0$.

By the definition, we have to prove that for every sequence $\{x_n\}$ such that $x_n \rightarrow 0$ as $n \rightarrow +\infty$ and $x_n \neq 0, \forall n$, we have:

$$f(x_n) \rightarrow 1 \quad \text{as } n \rightarrow +\infty$$

We have

$$f(x_n) = 1, \quad \forall n, \quad \text{hence } f(x_n) \rightarrow 1 \quad \text{as } n \rightarrow +\infty$$

Consequently $f(x) \rightarrow 1$ as $x \rightarrow 0$. In this case, therefore (unlike the previous example), we have:

$$\lim_{x \rightarrow 0} f(x) \neq f(0)$$

Reflecting on the two examples

- ▶ Think about the two examples just seen. In both cases the limit at a finite point of a certain function exists and is finite.
- ▶ In the first case, this limit coincides with the value of the function at the point considered; in the second case it does not.
- ▶ We therefore introduce the concept of continuity to distinguish these two cases.

Summary

Finite limit at a finite point:

- ▶ $f(x) \rightarrow \ell \in \mathbb{R}$ as $x \rightarrow c \in \mathbb{R}$
- ▶ $\sin x \rightarrow 0 = \sin 0$ as $x \rightarrow 0$ (comparison theorem with $|\sin x| \leq |x|$)
- ▶ The limit can also differ from the value $f(c)$, as for the function equal to 1 for $x \neq 0$ and to 0 at $x = 0$

Continuity

Introduction: continuity

- ▶ In this section we define **continuous functions**: those for which the limit at a point coincides with the value of the function at that point
- ▶ We will see the geometric interpretation of continuity and the **jump discontinuities**
- ▶ Continuity expresses the idea that small variations of x produce small variations of $f(x)$

Definition 13: Continuity

If $f : I \rightarrow \mathbb{R}$ and $c \in I$, we say that f is **continuous** at c if

$$\lim_{x \rightarrow c} f(x) = f(c)$$

We say that f is **continuous in I** if it is continuous at each point of I .

- ▶ The continuity property on an interval has a simple geometric interpretation:
“the graph of a function continuous on an interval can be drawn, on that interval, without lifting the pen from the paper”
- ▶ A function that is **not continuous** at a point c is called **discontinuous** at c .

Example

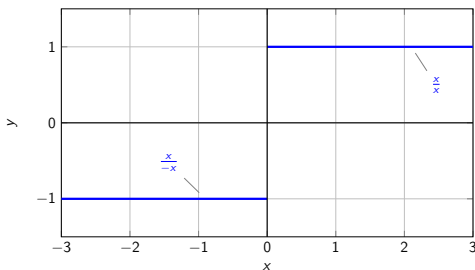
The function:

$$f(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is discontinuous at 0. The function $\sin x$ is continuous at 0.

Example

Consider the function: $f(x) = x/|x|$



In this case it is not possible to compute $f(0)$ and the function $f(x)$ is discontinuous at 0 since:

$$\lim_{x \rightarrow 0^+} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = -1$$

Jump discontinuity

Definition 14: Jump discontinuity point

We say that c is a **jump discontinuity point** of f when the right and left limits at c exist and are finite, but are different from each other. The jump is given by the difference of the limits:

$$\text{jump at } c = \lim_{x \rightarrow c^+} f(x) - \lim_{x \rightarrow c^-} f(x)$$

Example: the function

$$f(x) = \frac{x}{|x|}$$

has a jump discontinuity point at $x = 0$, with jump equal to 2.

Continuity from the right and from the left

- ▶ If one of the two limits, the right limit or the left limit (as $x \rightarrow c$), coincides with $f(c)$, we say that f is **continuous from the right or from the left**, respectively.
- ▶ Functions with jump discontinuities are well suited to model phenomena that undergo sudden changes.
- ▶ Continuous functions, on the other hand, owe their importance to the fact that

$$\lim_{x \rightarrow c} f(x) = f(c)$$

can be interpreted by saying that “if x is close to c ” then “ $f(x)$ is close to $f(c)$ ”, that is, small variations of x imply small variations of $f(x)$.

Summary

Continuity:

- ▶ f is continuous at c if $\lim_{x \rightarrow c} f(x) = f(c)$; continuous in I if it is so at every point of I
- ▶ Geometrically: the graph can be drawn without lifting the pen from the paper

Discontinuities:

- ▶ Jump discontinuity: right and left limits finite but different; example $x/|x|$ with jump 2
- ▶ Continuity from the right (left): the right (left) limit coincides with $f(c)$

Non-existence of the limit

Introduction: non-existence of the limit

- ▶ The limit of a function may also not exist:

$$\lim_{x \rightarrow c} f(x) \text{ does not exist}$$

- ▶ With the sequential definition it is enough to find **two sequences** tending to c along which the function tends to **two different limits**
- ▶ We see this for $\sin x$ as $x \rightarrow +\infty$ and for $\sin \frac{1}{x}$ as $x \rightarrow 0$

Example (1/2)

Let us prove that:

$$\lim_{x \rightarrow +\infty} \sin x \text{ does not exist}$$

To prove it, it is enough to find two sequences $\{x_n\}$ and $\{y_n\}$ such that $x_n \rightarrow +\infty$ and $y_n \rightarrow +\infty$ as $n \rightarrow +\infty$, with $\{\sin x_n\}$ and $\{\sin y_n\}$ tending to two different limits. We take:

$$x_n = n\pi \quad \text{and} \quad y_n = \frac{\pi}{2} + 2n\pi, \quad \forall n$$

consequently we have:

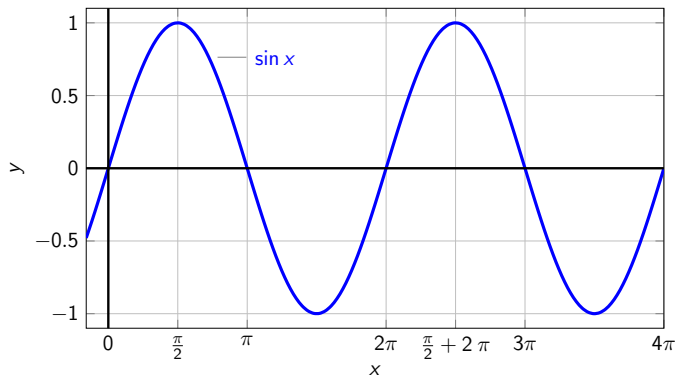
$$\{n\pi\} \rightarrow +\infty \quad \text{and} \quad \left\{\frac{\pi}{2} + 2n\pi\right\} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty$$

For these sequences we have

$$\sin x_n = 0 \quad \text{and} \quad \sin y_n = 1, \quad \forall n$$

Example (2/2)

and consequently the two sequences tend to two different limits. Hence the definition of limit is not satisfied and the limit does not exist.



Example (1/3)

Let us prove that:

$$\lim_{x \rightarrow 0} \sin \frac{1}{x} \text{ does not exist}$$

To prove it, it is enough to find two sequences $\{x_n\}$ and $\{y_n\}$ such that $x_n \rightarrow 0$, $y_n \rightarrow 0$ as $n \rightarrow +\infty$ and $x_n, y_n \neq 0, \forall n$, with $\left\{\sin \frac{1}{x_n}\right\}$ and $\left\{\sin \frac{1}{y_n}\right\}$ tending to two different limits. We take:

$$x_n = \frac{1}{n\pi} \quad \text{and} \quad y_n = \frac{1}{\frac{\pi}{2} + 2n\pi}, \quad \forall n > 1$$

consequently we have:

$$\left\{\frac{1}{n\pi}\right\} \rightarrow 0 \quad \text{and} \quad \left\{\frac{1}{\frac{\pi}{2} + 2n\pi}\right\} \rightarrow 0 \quad \text{as} \quad n \rightarrow +\infty$$

Example (2/3)

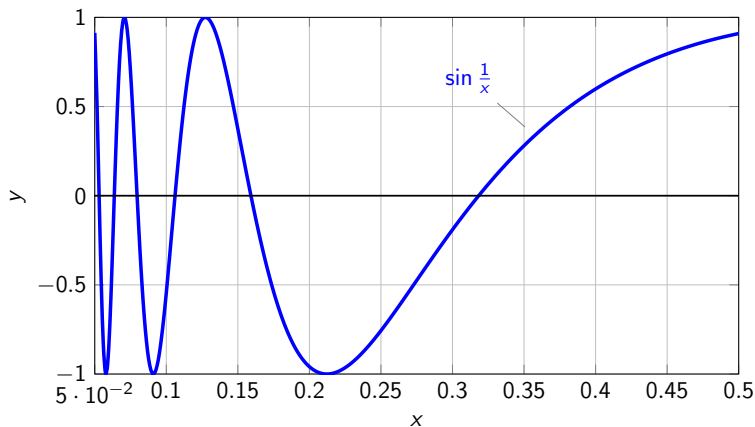
For these sequences we have

$$\sin \frac{1}{x_n} = 0 \quad \text{and} \quad \sin \frac{1}{y_n} = 1, \quad \forall n > 1$$

and consequently the two sequences tend to two different limits.
Hence the definition of limit is not satisfied and the limit does not exist.

The function has infinitely many oscillations in a finite space, so it is not possible to draw its graph on an interval $(0, b]$ with $b \in \mathbb{R}_+$.

Example (3/3)



Non-existence of the limit:

- ▶ To prove that $\lim_{x \rightarrow c} f(x)$ does not exist it is enough to find two sequences $x_n, y_n \rightarrow c$ (with $x_n, y_n \neq c$) such that $f(x_n)$ and $f(y_n)$ tend to different limits
- ▶ $\lim_{x \rightarrow +\infty} \sin x$ does not exist: $x_n = n\pi$, $y_n = \frac{\pi}{2} + 2n\pi$
- ▶ $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ does not exist: infinitely many oscillations near 0

Final summary (1/2)

- ▶ **Sequential and topological definition** of limit: they are equivalent; the limit, if it exists, is unique
- ▶ **Finite limit at infinity**: horizontal asymptote; limits from above or from below
- ▶ **Infinite limit at infinity**: possible oblique asymptote
 $y = mx + q$, with $m = \lim_{x \rightarrow \infty} \frac{f(x)}{x}$ and $q = \lim_{x \rightarrow \infty} (f(x) - mx)$
- ▶ **Infinite limit at a finite point**: vertical asymptote; right and left limits

Final summary (2/2)

- ▶ **Finite limit at a finite point:** the limit may or may not coincide with $f(c)$
- ▶ **Continuity:** $\lim_{x \rightarrow c} f(x) = f(c)$; jump discontinuities; continuity from the right and from the left
- ▶ **Non-existence of the limit:** two sequences tending to c whose images have different limits